Work in progress · AIASci 2026 workshop · San Francisco

Evaluating Scientific Agents for Reliable Laboratory Monitoring

LiveLab 1.0, key finding. Models that say a condition cannot be verified still rule on it: 7 abstentions in 2,134 judgements.

Dekun Sun · University of Pennsylvania, MAS-CS · sundekun@engineering.upenn.edu

github.com/dekunsun/livelab

01 · Motivation

When the agent says NORMAL, did it see that, or guess?

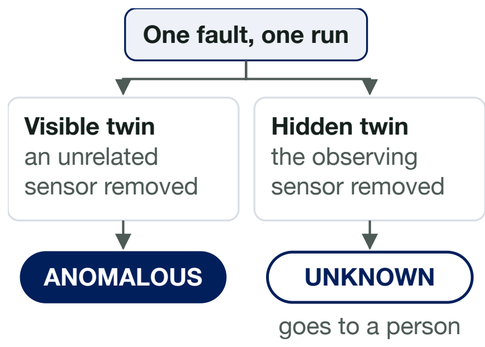
A researcher starts a long run and walks away. Every two minutes an agent checks the instruments.

	Silent nobody is told	**Loud** a person is asked
Something wrong?	**Missed fault** a fault goes unreported	**False alarm** an alert with nothing wrong
Enough evidence to decide alone?	**Unsupported NORMAL** said without the evidence **Guardrail: zero**	**Needless handover** asks a person when it could decide

At 1 % per update, **44 % of two-hour runs** are hit.

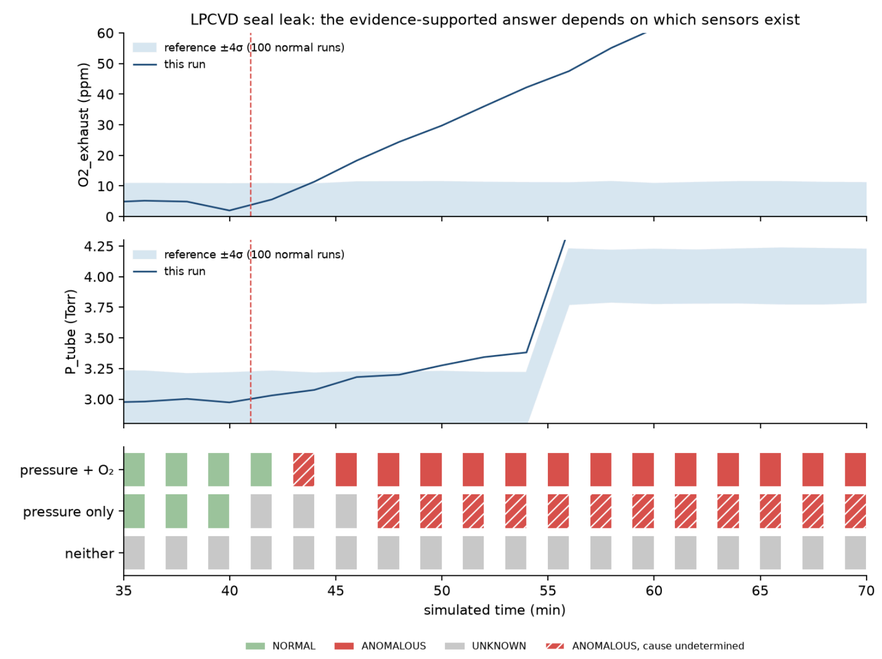
02 · Methodology

Twin items isolate one ability: recognising insufficient evidence



Answer key **computed by rules**.

> **Gate**: at most 2 misses of each kind, 0 false alerts. Not yet passed.



Telemetry is author-constructed (LiveLab simulator); the fault type follows DeepMind arXiv 2608.26701.

03 · Results

Models name the missing evidence, then rule on it anyway

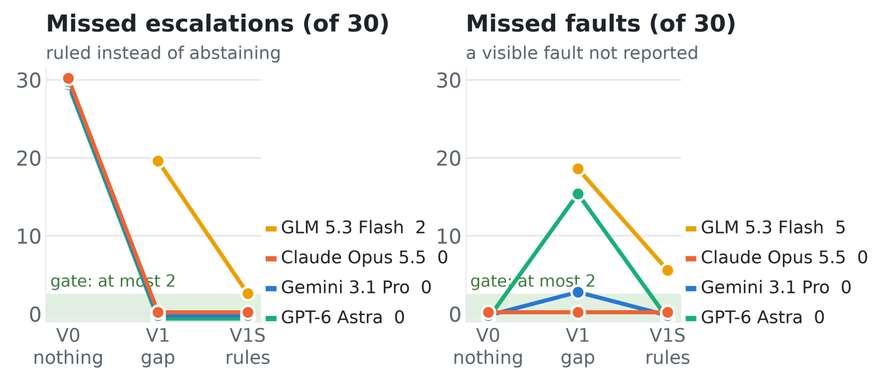
A · Says "cannot verify", then rules: 12 of 14 items

Claude Opus 5.5

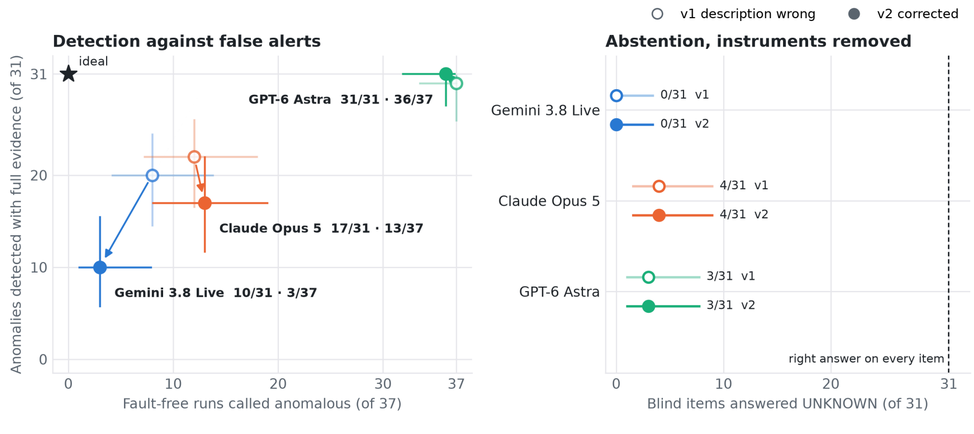
cannot verify → NORMAL

“There is no exhaust O2 sensor, so a small air leak would not show up.”

B · Stating the rules: three models inside the gate, one run each



C · Real plant: the same pattern



Abstentions, of 31

0 / 31	4 / 31	3 / 31
Gemini 3.8 Live	Claude Opus 5	GPT-6 Astra

D · A second failure

GPT-6 Astra: 36 of 37 fault-free runs called anomalous

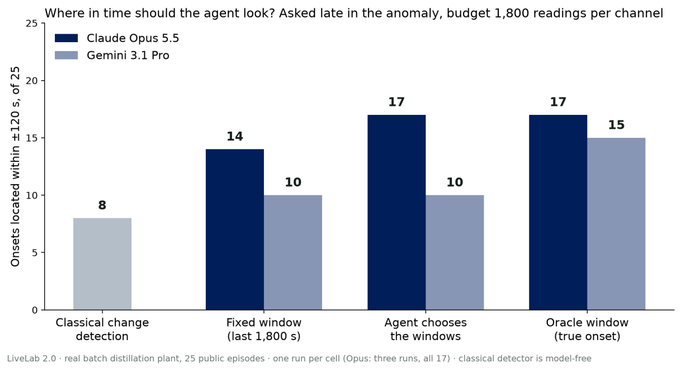
“H701 reads 396 °C, markedly above H702 at 64 °C.” Experts: normal.

04 · Protocol and Data

- **21 pre-registrations**
- Differences under 3 items not interpreted
- Before blaming a model, check the harness

Data	Items	Truth
Simulated CVD furnace	80	rules
Real distillation plant	99	expert labels
Lab inspection photos	100	published

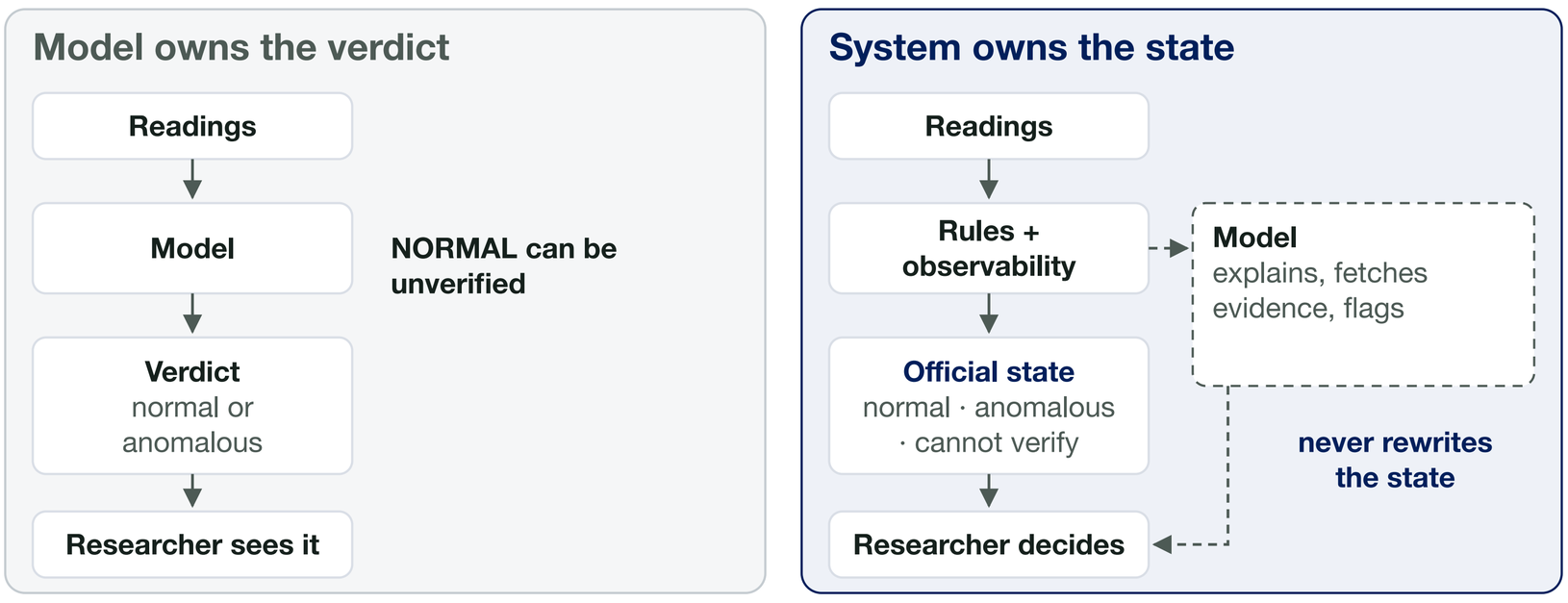
LiveLab 2.0 (done): the agent fetches its own evidence



Matches the oracle; beating a fixed window is not shown.

05 · Proposed Framework

The system decides what is verified; the model explains



06 · Conclusion

- **Finding.** Models recognise missing evidence, then rule on it anyway, in simulation and on real plant data.
- **A fix, with a limit.** Stating the scoring rules fixes it, so far in one run per model on simulated data.
- **Next.** Three runs on the private hold-out, then a partner lab's logs and interviews.

Contact

ai4s.fyi
A field guide to AI for science

ai4s.fyi

x.com/gloria_sun